

NLP

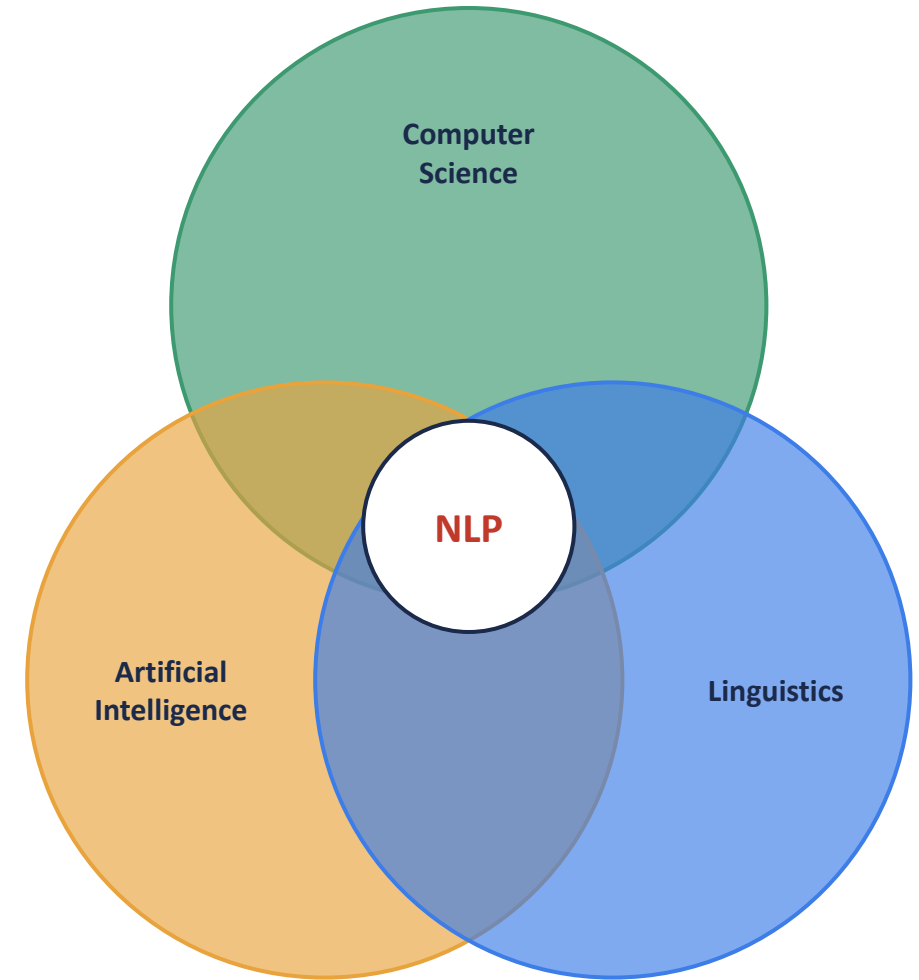
Introduction to Natural Language Processing

Teaching how machines read, understand, and generate human language

What is NLP?

Natural Language Processing is a subfield of linguistics, computer science, and artificial intelligence concerned with the interactions between computers and natural language — in particular, how to program computers to process and analyze large amounts of natural language data.

NLP sits at the intersection of three fields — it borrows structure from Computer Science, reasoning from Artificial Intelligence, and its subject matter from Human Language.



What is NLP?

Natural Language Processing is the branch of Artificial Intelligence that gives computers the ability to read, understand, interpret, and generate natural language — in both text and speech.



A bridge between two very different worlds:

“

I saw the man on the hill with a telescope.

Who has the telescope?

This single sentence has multiple valid readings — the kind of ambiguity NLP systems must resolve.

What is a Natural Language?

The starting point for understanding NLP

In neuropsychology, linguistics, and the philosophy of language, a natural language (or ordinary language) is any language that has evolved naturally in humans through use and repetition, without conscious planning or premeditation.

Natural languages can take different forms, such as speech or signing. They are distinguished from constructed and formal languages, such as those used to program computers or study logic.

Natural Language

English, Hindi, Tamil — evolved organically

Formal Language

Python, Logic notation — designed deliberately

Why is NLP Needed?

Understanding the real-world forces driving Natural Language Processing

The Data Explosion

Most of the world's data is text — and it's unusable without NLP

80%+

of all enterprise data is unstructured text

*emails · chat logs · reviews · reports · medical notes ·
legal filings*



Social & Messaging

Billions of posts, comments, short videos, memes and messages generated daily



Business Documents

Contracts, reports, tickets, and internal communications



Web Content

Articles, product listings, reviews across every domain



Speech & Audio

Calls, meetings, and voice interactions needing transcription

Structured databases and manual review cannot keep pace — NLP is the only scalable path to extracting meaning from this volume.

Humans Speak Language. Machines Speak Logic.

NLP exists to bridge this fundamental mismatch

How Humans Communicate

- Ambiguous — words carry multiple meanings
- Context-dependent — meaning shifts with situation
- Ever-evolving — slang, idioms, new usage
- Implicit — much is left unsaid, inferred



How Machines Operate

- Precise — operates on fixed, unambiguous symbols
- Structured — expects defined fields and formats
- Rule-consistent — no tolerance for inconsistency
- Explicit — requires everything stated directly

Manual Processing Doesn't Scale

What a human team can read vs. what businesses actually generate

A person reading full-time



~200–300 documents / day

A mid-size company's daily support tickets



10,000+ / day

Global social media posts

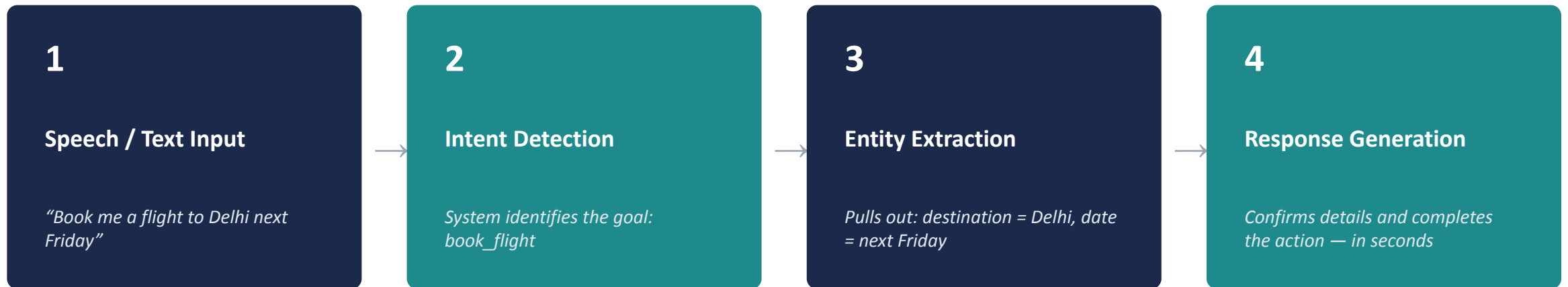


500 million+ / day

NLP automates triage, classification, and summarization — closing a gap no team can close by hiring alone.

Real-Time Systems Demand Language Understanding

Voice assistants, chatbots, and live support can't wait for a human to interpret



All of this happens in under a second — powered entirely by NLP components working together.

Why NLP is Needed — In Short

- Most data in the world is unstructured text — NLP is the only scalable way to use it
- Machines process logic; humans communicate in ambiguous, contextual language
- Manual review cannot match the volume of content generated today
- Real-time systems (voice, chat) require instant language understanding
- Translation, search, summarization, and sentiment analysis all depend on it
- Without NLP, most of the digital services we rely on daily wouldn't exist

Next: How NLP systems are actually built — the pipeline in detail

Real-World Applications

NLP is already embedded in tools used every day

1

Contextual Advertisements

Matching ad content to the meaning of the page or query

2

Email

Spam filtering and smart-reply suggestions

3

Social Media

Removing adult/harmful content and mining public opinion

4

Search Engines

Understanding query intent, not just keywords

5

Chatbots

Automated, conversational customer interaction

Applications of NLP

NLP powers tools we use every single day



Machine Translation

Google Translate, DeepL



Virtual Assistants

Siri, Alexa, Google Assistant



Sentiment Analysis

Product & social media reviews



Chatbots

Customer support automation



Search Engines

Query understanding & ranking



Spam / Fraud Detection

Email and message filtering



Text Summarization

News digests, meeting notes



Speech Recognition

Voice-to-text transcription



Grammar Correction

Grammarly, Word editor

1

Text Classification

Assigning categories or labels to text

2

Named Entity Recognition

Detecting people, places, organizations

3

Sentiment Analysis

Determining emotional tone of text

4

Machine Translation

Converting text between languages

5

Question Answering

Producing answers from context

6

Text Summarization

Condensing text while keeping meaning

Why is Language Hard for Machines?

Ambiguity exists at every level of language



Lexical Ambiguity

“Bank” — a river bank or a financial bank?



Syntactic Ambiguity

“I saw the man with a telescope” — who holds it?



Referential Ambiguity

“The trophy didn’t fit the suitcase because it was too big” — what is “it”?



Contextual / Pragmatic

“Can you pass the salt?” is a request, not a question about ability

Challenges in NLP

Why language remains genuinely hard for machines to process

Ambiguity

"I saw the boy on the beach with my binoculars." — who has the binoculars?

Colloquialisms & Slang

"Piece of cake", "pulling your leg" — meaning isn't literal

Irony, Sarcasm & Tone

"That's just what I needed today!" — said after something goes wrong

Creativity

Poems, dialogue, and scripts bend or break normal grammar rules

Contextual Words

"I ran to the store because we ran out of milk." — same word, two meanings

Synonyms

Many words share meaning but differ in tone, formality, or nuance

Spelling Errors

Typos and non-standard spelling break simple word-matching

Open Challenges in NLP



Low-Resource Languages

Limited labeled data for many world languages



Common-Sense Reasoning

Understanding implied real-world knowledge



Sarcasm & Irony

Intended meaning opposite of literal words

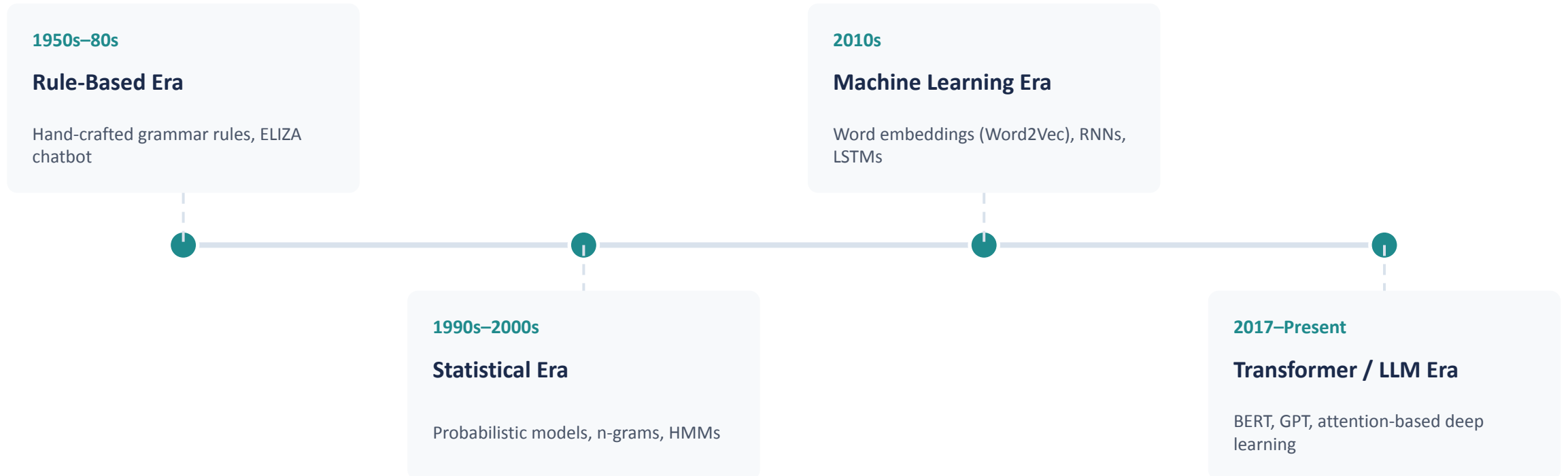


Computational Cost

Large models require heavy compute resources

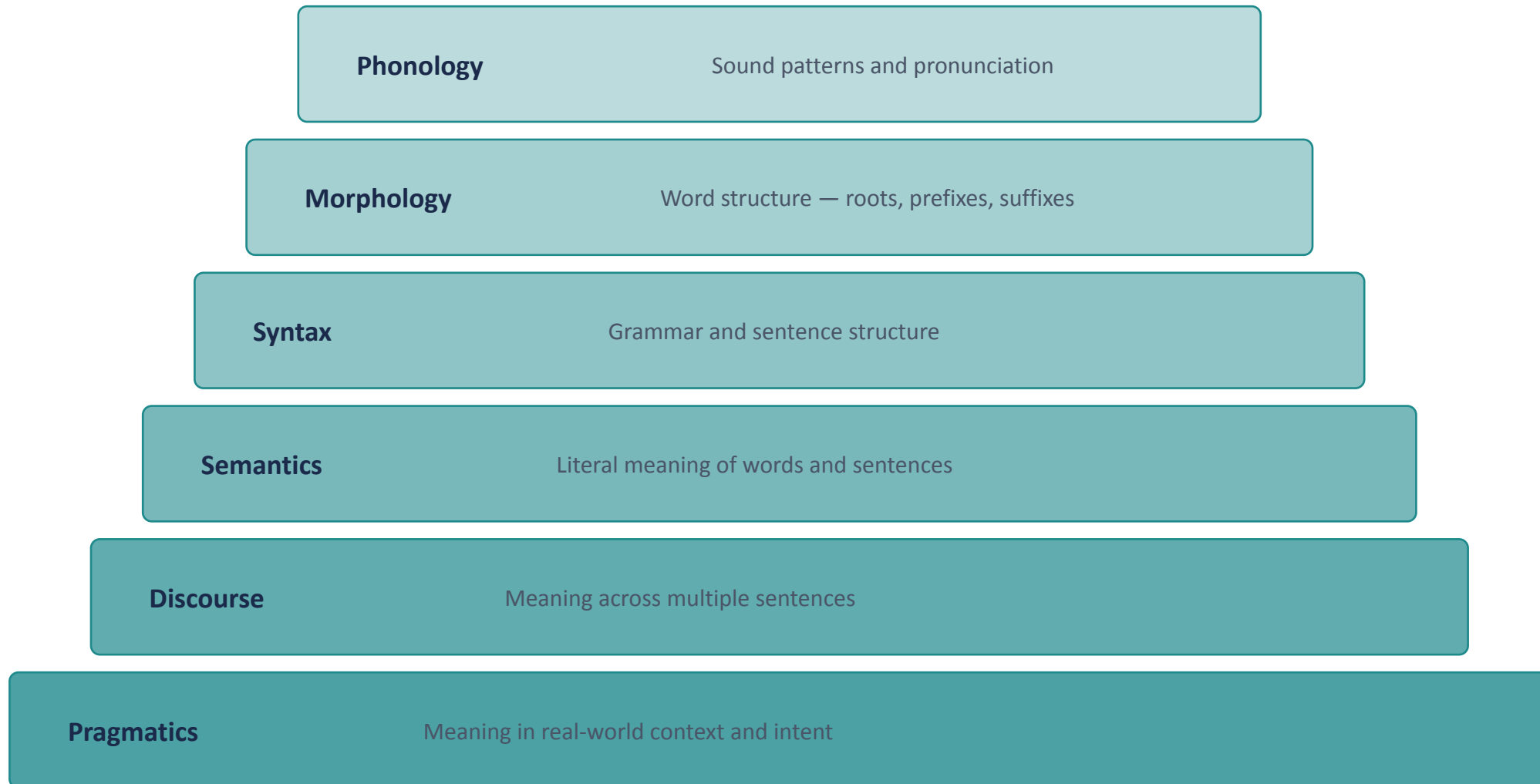
Evolution of NLP

From hand-written rules to large language models



Levels of Linguistic Analysis

NLP systems process language through layered levels of understanding



A Typical NLP Pipeline

Raw text moves through several preprocessing stages before analysis



Example

Input: “The students are running quickly to class.”

Tokens: [The, students, are, running, quickly, to, class] → Lemmas: [run, quick, class] → POS: [DT, NNS, VBP, VBG, RB, TO, NN]

Approaches to NLP



1950s–80s

Rule-Based

Hand-written linguistic rules and grammars. Predictable but brittle and hard to scale.



1990s–2000s

Statistical

Probabilities learned from large text corpora (n-grams, HMMs, CRFs).



2000s–10s

Machine Learning

Feature-based classifiers — Naive Bayes, SVM — trained on labeled data.



2017–Present

Deep Learning

Neural networks (RNN, LSTM, Transformers) learn representations directly from data.

Popular NLP Tools & Libraries

A

NLTK

Classic Python library for teaching & prototyping

B

spaCy

Fast, production-ready NLP pipeline

C

**Hugging Face
Transformers**

Pretrained models — BERT, GPT, T5

D

Gensim

Topic modeling & word embeddings

E

Stanford CoreNLP

Java-based linguistic analysis toolkit

F

TextBlob

Simple API for common NLP tasks

Key Takeaways

- NLP enables machines to understand and generate human language
- Language is inherently ambiguous — lexical, syntactic, and pragmatic
- A standard pipeline: tokenize → normalize → tag → parse → extract meaning
- The field evolved from rule-based systems to statistical models to today's deep learning and transformers
- NLP already powers translation, assistants, search, and chatbots — and is rapidly growing

Next class: Text Preprocessing & Tokenization in Depth

Assignment 1- NLP landscape from 1960s to 2026 (About NLP and its history)